EPR pre-generation on ten benchmarks

ODG, CGP, ACGP and compiler-driven Dynamic, memories per core $K = 1$ to ∞

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Chapter 1

Setup

1.1 Architecture and workloads

- 4×4 mesh of 16 cores; each core has 16 data slots, 14 occupied at the start and 2 kept free for migration, so circuits have $16 \times 14 = 224$ logical qubits.
- Benchmarks: ten MQT Bench 2.3.0 circuits at 224 qubits (`benchmarks/`, verified by `benchmarks/verify.py`). Each is mapped once (T-HQA placer, OEE initial placement, mapping seed 42) and routed with XY and with history routing (history counts the route visits of all preceding layers, i.e. cumulative history, routing capacity 8); every strategy and every K sees the same frozen trace.
- Physics (`config/final_config/grid_4x4_ace_3.json`): memory efficiency 0.8, memory fidelity 0.95, coherence 1 s, detector efficiency 0.95. The classical delay is set by the topology code to about 0.1 ms between neighbours.

1.2 Strategies and policy

ODG generates every pair on demand. CGP and ACGP generate pairs continuously into a cache. Dynamic is compiler driven: the planner chooses, before the run, the layer in which each pair is started. Everything except K and the memory split is the default policy of the code (branch `history-algo-update`, `ace/compiler/settings.py`):

Setting	Value
Layer basis, lead, lookahead, age limit	active layers; 1, 2, 10 layers
Generation budget	8 launches per core per active layer
Preparation priority	trace order (`trace`)
Demand guard / refill on release	on / on
Admission	per-qubit, event driven, ready pairs first, longer chains first
Empty-layer gap	10 ns
Compiler reservation, pair use	50 ms, handoff
Physics seeds	60 and 61

Table 1.1: Default policy used in every run.

1.3 Memories per core and the split

K (memories per core)	1	2	3	4	6	8	∞ (64)
Dynamic may use	1	2	3	4	6	6	64
CGP / ACGP cache ($\lfloor 0.75K \rfloor$)	0	1	2	3	4	6	48

Table 1.2: Pre-generation share of the K memories. At $K = 8$ six memories are reserved for pre-generation. CGP/ACGP use $\lfloor 0.75K \rfloor$ because caching in all K memories starves requests (smoke test: 150–1060 ms per hop at $K = 1$–4); at $K = 1$ they therefore equal ODG.

1.4 Metrics

A *hop* is one request between neighbouring cores; a *chain* is all hops of one qubit in one layer.

Latency mean time per hop from eligibility to EPR delivery, including any wait for a free memory.

Fidelity mean fidelity of the delivered pairs (not a circuit fidelity).

Coverage share of hops that found a pair already made: generation phase under 0.35 ms, i.e. only the reservation handshake. Same definition for every strategy. Chain full coverage: every hop of the chain ready.

Utilisation pairs used divided by pairs generated. **Wastage** = 100− utilisation: pairs that expired or were never used. ODG makes no pairs in advance, so both are not applicable to it.

Bars are the mean of the two seeds; whiskers span the two seeds.

Chapter 2

Results

2.1 QFT, XY routing

11076 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

2.1.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic (∞) uses 64, which no core ever fills.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.782	4.60	0.9498	0.0	0.0	–	–
CGP	0.970	3.25	0.9424	77.9	60.1	4.24	95.8
ACGP	0.976	3.30	0.9421	77.6	60.0	4.39	95.6
Dynamic (8)	0.747	2.25	0.9439	76.8	64.5	1.23	98.8
Dynamic (∞)	0.616	1.65	0.9442	84.0	71.9	1.18	98.8

Table 2.1: QFT, XY routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

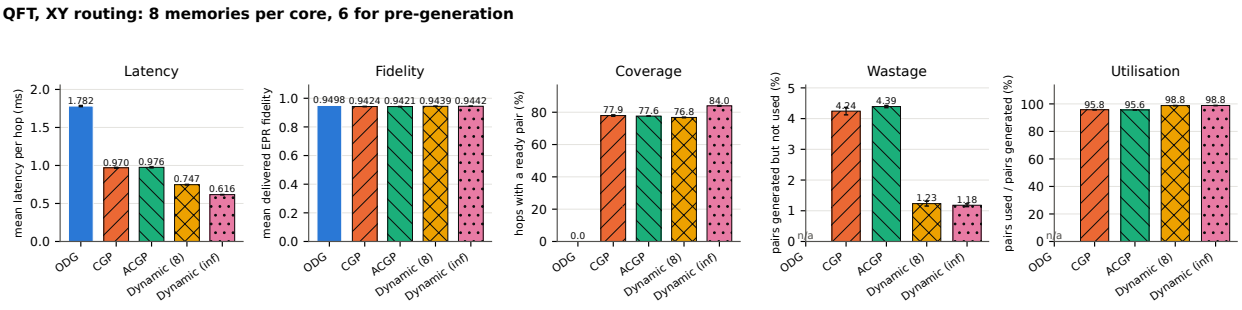


Figure 2.1: QFT, XY routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.1.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the K sweep; Dynamic (∞) as above.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.887	4.80	0.9498	0.0	0.0	–	–
CGP	2.331	8.95	0.9401	63.0	41.7	5.87	94.1
ACGP	2.325	8.95	0.9402	63.4	43.9	5.76	94.2
Dynamic (4)	1.295	4.70	0.9425	63.7	54.1	3.89	96.1
Dynamic (∞)	0.616	1.65	0.9442	84.0	71.9	1.18	98.8

Table 2.2: QFT, XY routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

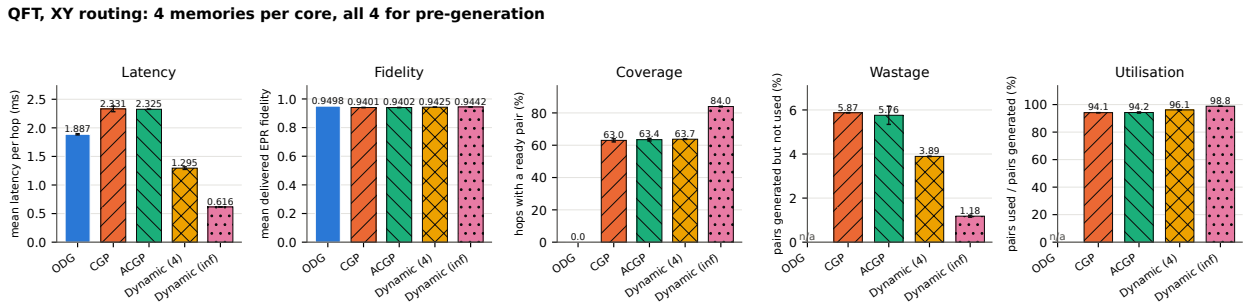


Figure 2.2: QFT, XY routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.1.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

Latency	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	6.275	2.816	2.109	1.887	1.788	1.782	1.782
CGP	6.275	4.177	2.885	2.331	1.237	0.970	0.457
ACGP	6.275	4.118	2.974	2.325	1.247	0.976	0.458
Dynamic	9.041	3.528	1.877	1.295	0.866	0.747	0.616

Table 2.3: QFT, XY routing: latency (mean latency per hop (ms)) for every K.

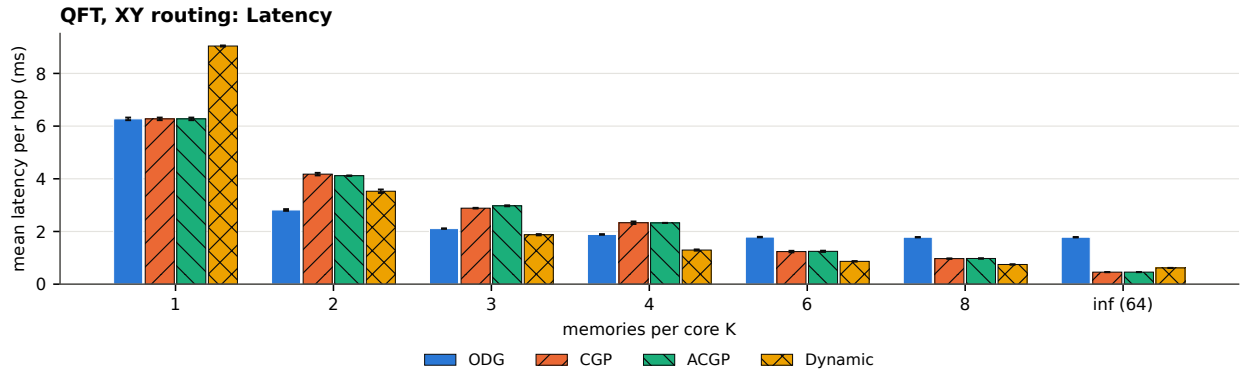


Figure 2.3: QFT, XY routing: latency for every K.

Fidelity	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498
CGP	0.9498	0.9441	0.9415	0.9401	0.9427	0.9424	0.9422
ACGP	0.9498	0.9441	0.9414	0.9402	0.9425	0.9421	0.9422
Dynamic	0.9486	0.9440	0.9429	0.9425	0.9432	0.9439	0.9442

Table 2.4: QFT, XY routing: fidelity (mean delivered EPR fidelity) for every K.

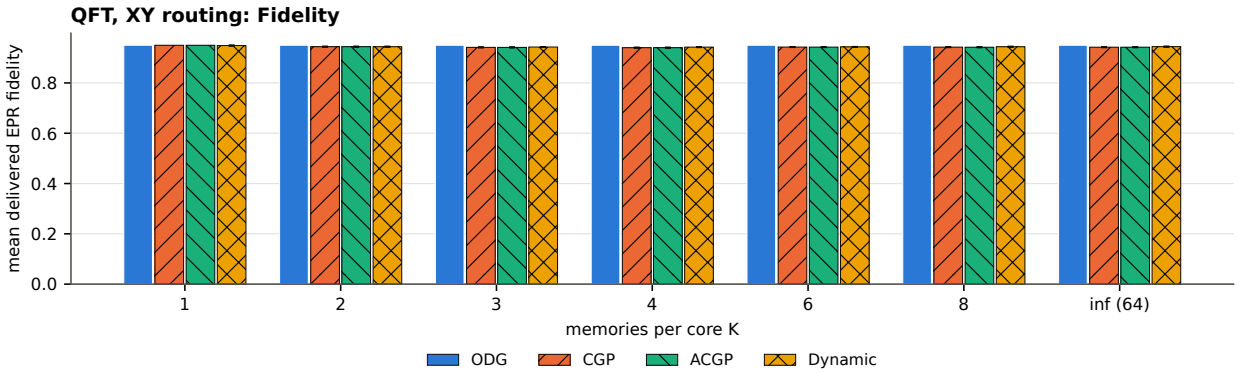


Figure 2.4: QFT, XY routing: fidelity for every K.

Coverage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CGP	0.0	26.8	50.5	63.0	64.2	77.9	99.3
ACGP	0.0	27.0	49.0	63.4	63.9	77.6	99.1
Dynamic	8.9	36.3	52.4	63.7	76.8	76.8	84.0

Table 2.5: QFT, XY routing: coverage (hops with a ready pair (%)) for every K.

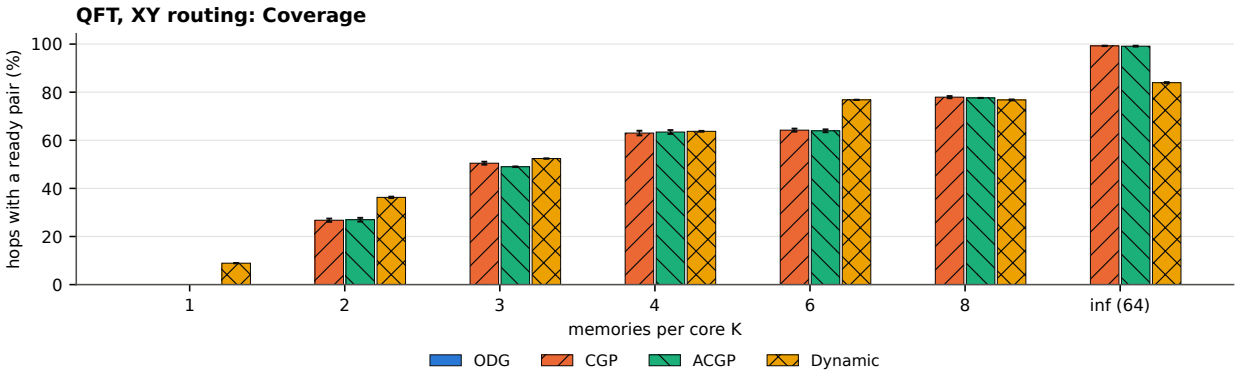


Figure 2.5: QFT, XY routing: coverage for every K.

Wastage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	7.08	5.66	5.87	4.34	4.24	13.73
ACGP	–	7.04	6.13	5.76	4.49	4.39	13.73
Dynamic	60.80	14.59	6.08	3.89	2.05	1.23	1.18

Table 2.6: QFT, XY routing: wastage (pairs generated but not used (%)) for every K.

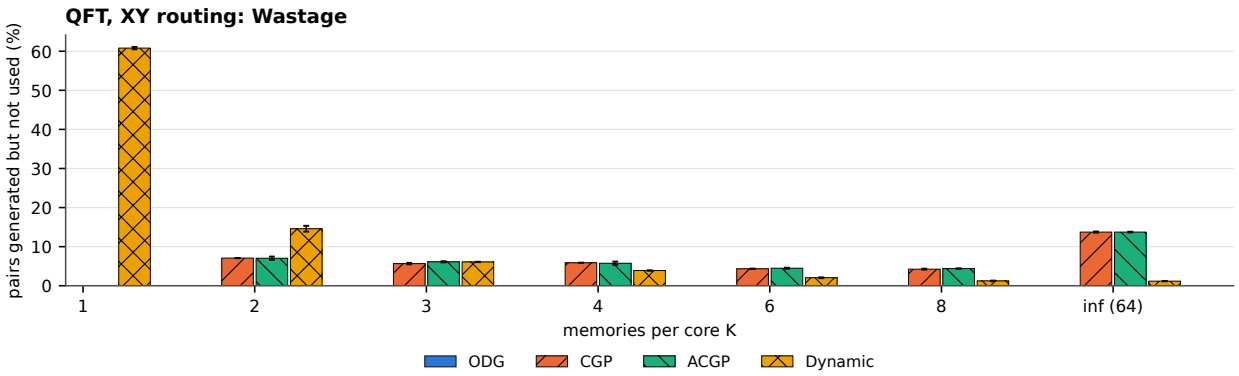


Figure 2.6: QFT, XY routing: wastage for every K.

Utilisation	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	92.9	94.3	94.1	95.7	95.8	86.3
ACGP	–	93.0	93.9	94.2	95.5	95.6	86.3
Dynamic	39.2	85.4	93.9	96.1	98.0	98.8	98.8

Table 2.7: QFT, XY routing: utilisation (pairs used / pairs generated (%)) for every K.

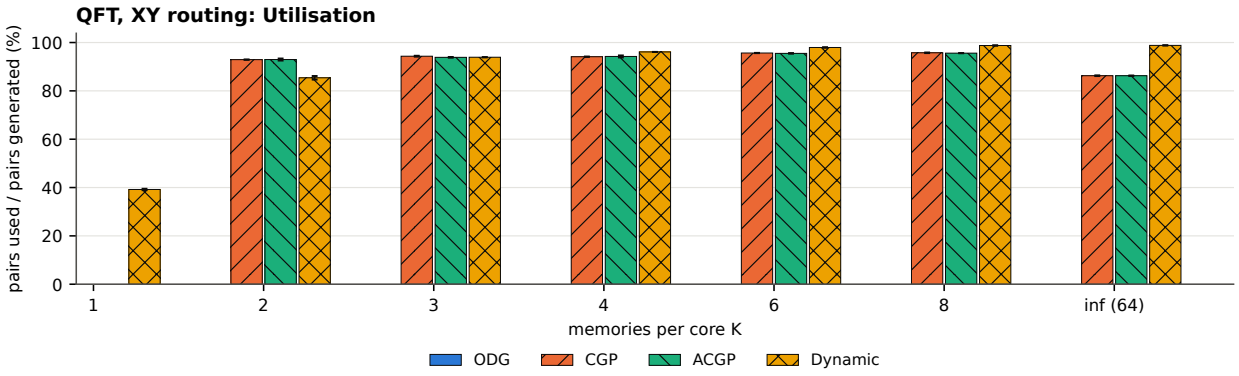


Figure 2.7: QFT, XY routing: utilisation for every K.

2.2 QFT, history routing

11076 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

2.2.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic (∞) uses 64, which no core ever fills.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.773	4.40	0.9498	0.0	0.0	–	–
CGP	0.966	3.10	0.9422	73.8	52.6	4.97	95.0
ACGP	0.917	2.95	0.9425	76.6	57.4	4.34	95.7
Dynamic (8)	0.693	2.00	0.9442	79.6	65.7	1.40	98.6
Dynamic (∞)	0.584	1.45	0.9450	85.6	72.4	1.46	98.5

Table 2.8: QFT, history routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

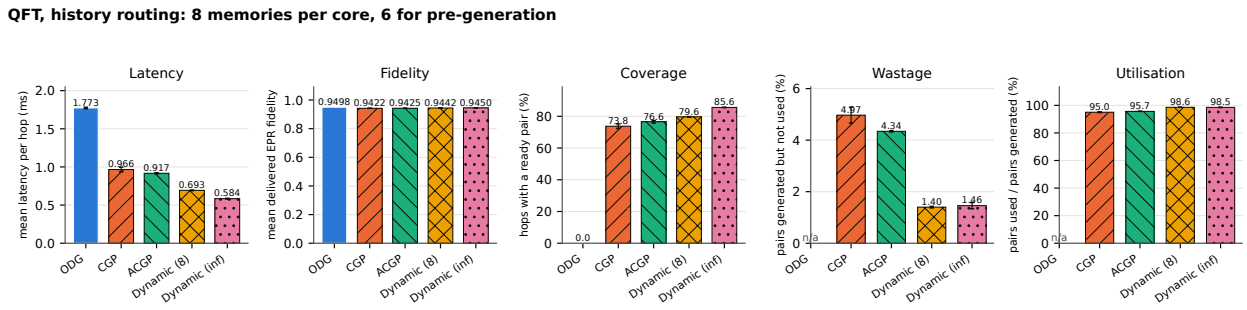


Figure 2.8: QFT, history routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.2.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the K sweep; Dynamic (∞) as above.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.824	4.60	0.9498	0.0	0.0	–	–
CGP	2.192	8.10	0.9413	63.5	40.3	5.37	94.6
ACGP	2.194	8.14	0.9412	62.3	38.3	5.44	94.6
Dynamic (4)	1.199	4.20	0.9423	65.8	55.5	4.45	95.6
Dynamic (∞)	0.584	1.45	0.9450	85.6	72.4	1.46	98.5

Table 2.9: QFT, history routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

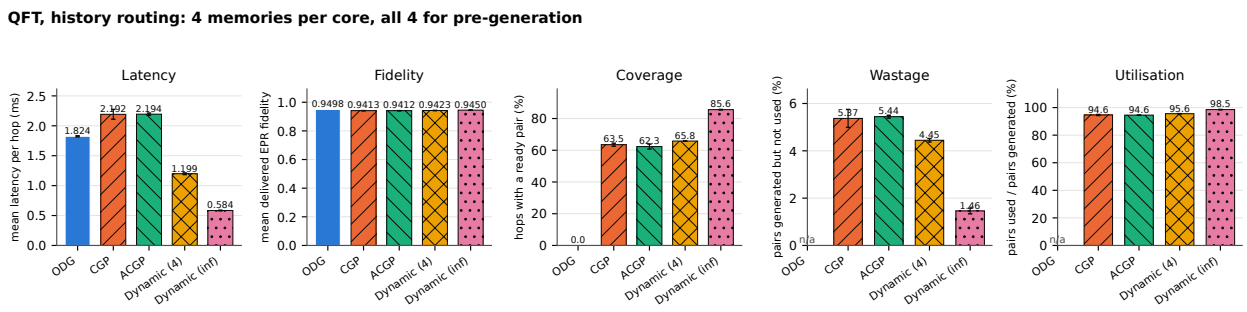


Figure 2.9: QFT, history routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.2.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

Latency	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	5.820	2.554	1.963	1.824	1.776	1.773	1.773
CGP	5.820	3.739	2.767	2.192	1.112	0.966	0.457
ACGP	5.820	3.748	2.696	2.194	1.113	0.917	0.455
Dynamic	8.812	3.153	1.749	1.199	0.797	0.693	0.584

Table 2.10: QFT, history routing: latency (mean latency per hop (ms)) for every K.

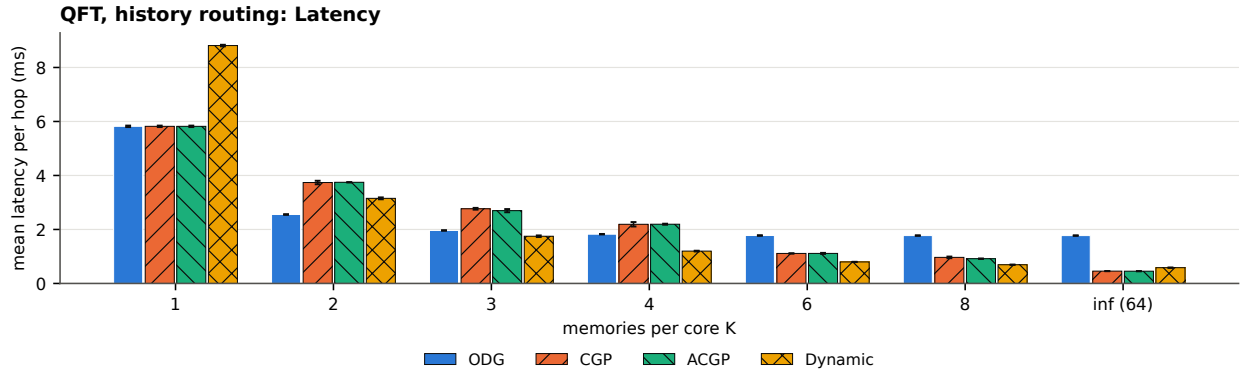


Figure 2.10: QFT, history routing: latency for every K.

Fidelity	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498
CGP	0.9498	0.9452	0.9427	0.9413	0.9433	0.9422	0.9417
ACGP	0.9498	0.9449	0.9426	0.9412	0.9433	0.9425	0.9418
Dynamic	0.9487	0.9442	0.9427	0.9423	0.9439	0.9442	0.9450

Table 2.11: QFT, history routing: fidelity (mean delivered EPR fidelity) for every K.

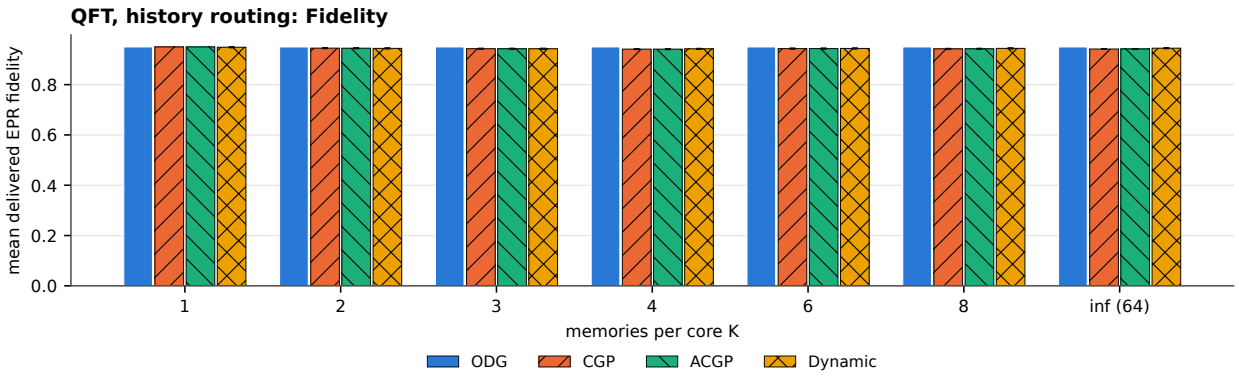


Figure 2.11: QFT, history routing: fidelity for every K.

Coverage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CGP	0.0	27.2	47.8	63.5	64.2	73.8	99.2
ACGP	0.0	28.2	49.7	62.3	65.3	76.6	99.5
Dynamic	8.4	36.4	53.8	65.8	79.2	79.6	85.6

Table 2.12: QFT, history routing: coverage (hops with a ready pair (%)) for every K.

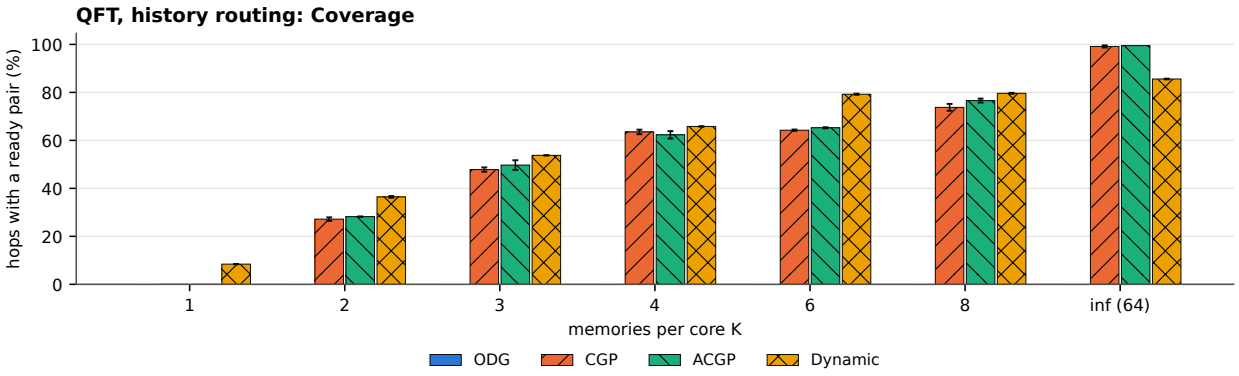


Figure 2.12: QFT, history routing: coverage for every K.

Wastage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	6.23	5.85	5.37	3.96	4.97	14.14
ACGP	–	6.07	5.58	5.44	3.74	4.34	13.66
Dynamic	63.26	14.98	6.07	4.45	2.25	1.40	1.46

Table 2.13: QFT, history routing: wastage (pairs generated but not used (%)) for every K.

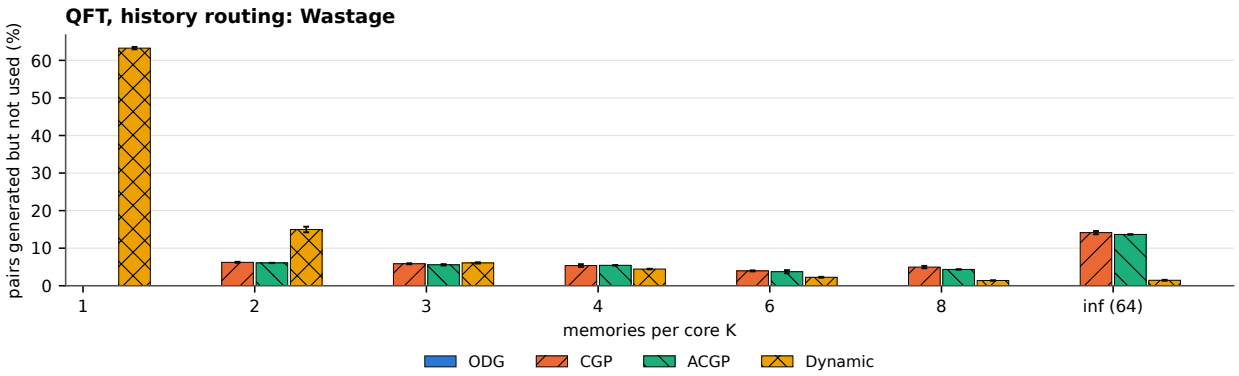


Figure 2.13: QFT, history routing: wastage for every K.

Utilisation	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	93.8	94.1	94.6	96.0	95.0	85.9
ACGP	–	93.9	94.4	94.6	96.3	95.7	86.3
Dynamic	36.7	85.0	93.9	95.6	97.7	98.6	98.5

Table 2.14: QFT, history routing: utilisation (pairs used / pairs generated (%)) for every K.

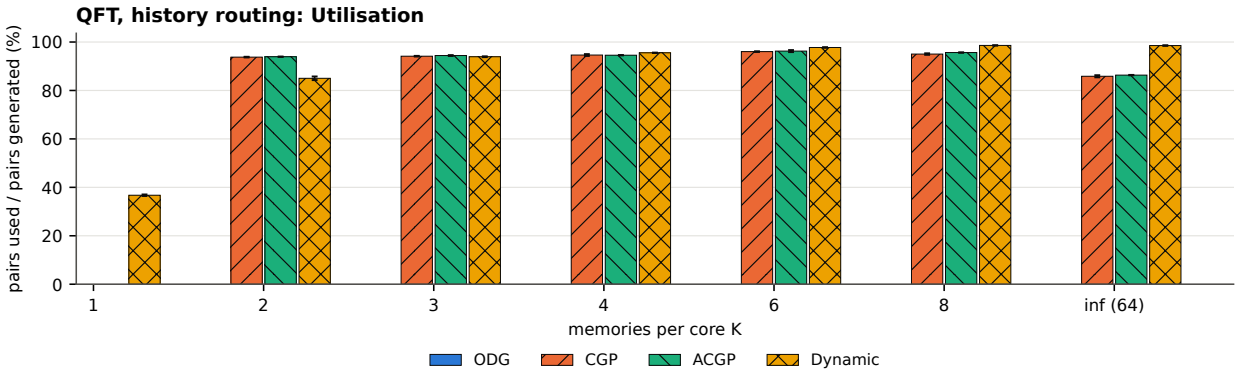


Figure 2.14: QFT, history routing: utilisation for every K.